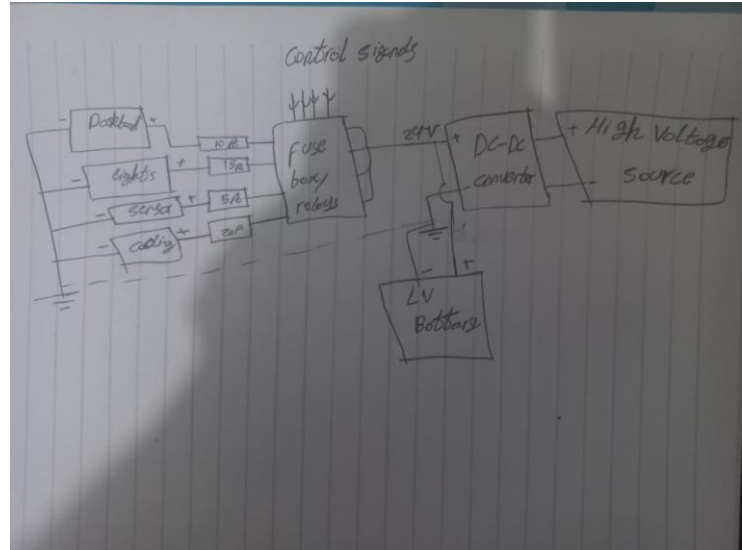


Architecture

In an electric vehicle there are mainly two power systems high voltage (HV) and low voltage (LV). The HV system is responsible for moving the vehicle powering the motors. They usually have a voltage anywhere from 200V to 800V. The LV system is responsible for powering all the rest of the vehicle



components which don't require this much voltage. The LV source is usually 12V but recently there has been a shift to use 48V as it is more efficient. The main way that these systems work is by using a dc-dc converter which converts the HV source to a LV source which is used to charge a battery. This battery is then used to power the LV system. This setup isolates the two systems for more protection. The output of the battery is then connected to power all the LV system with the help of a fuse box and control signals that determine which element to power and how much power should go there. All of the negatives of the battery and components are connected to the chases of the vehicle to act as a common ground.

Relays have two main benefits. The first they isolate high voltage noise power lines used to power loads from the low voltage signals to control the loads. The second is to protect the driving circuit as most loads require more current than what an ecu or a small physical switch can handle. Fuses are used to protect the wires and other components from over current. They are designed to be the first part of the circuit to fail when there is a surge in current so that no other parts will be damaged.

In manufacturing wires should be labeled with the same id as in the schematic. In addition there must be a colored coding to prevent mixed polarity. For routing the main idea is to account for vibration if the wire is too tight vibration will cause it to fall. If the wire isn't fixed it will flop around and might short circuit with the chassis. for protection we can use heat shielding and conduits. clean grounding is important for signal integrity and EMI shielding. For troubleshooting check the voltage in different location in the circuit from the power to the ground of the circuit until you identify where the problem is.

Fuse justification

For the fuses I used $I = \frac{P}{V}$ which for a voltage of 12 V gets following values

Power (Watt)	Fuse Rating (Amp)
60	5
120	10
180	15
240	20

Explanation of the simulated branch

- why the relay is used instead of driving the load directly from the switch

To drive the load directly the switch must have be able to sustain the relatively high current that the load requires which will not always be possible for example for an ecu to drive any load with more than a few milliamps will damage it.

- why a relay driver stage may be needed

To reduce the driving current even more, so that the ecu can power the coil with no risk.

- why the fuse is placed before the load

To protect the wires from overheating.

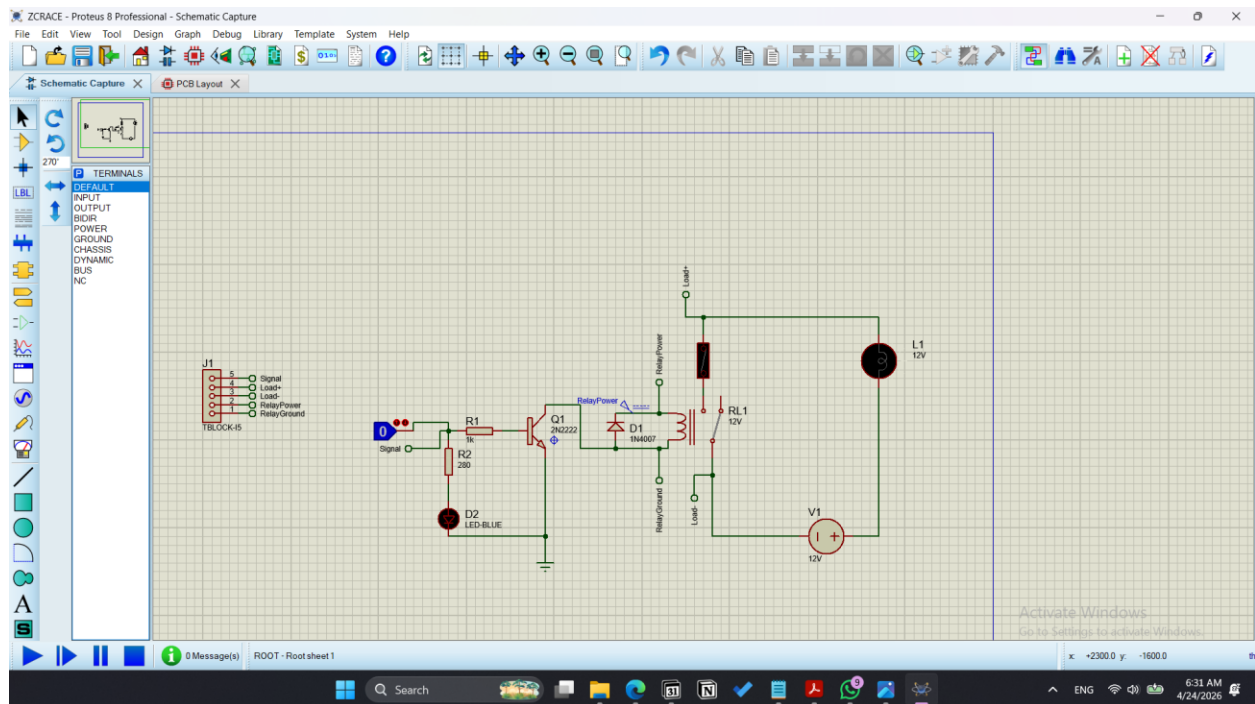
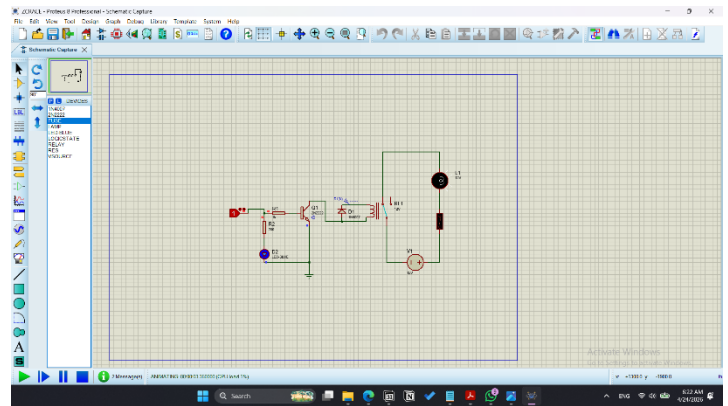
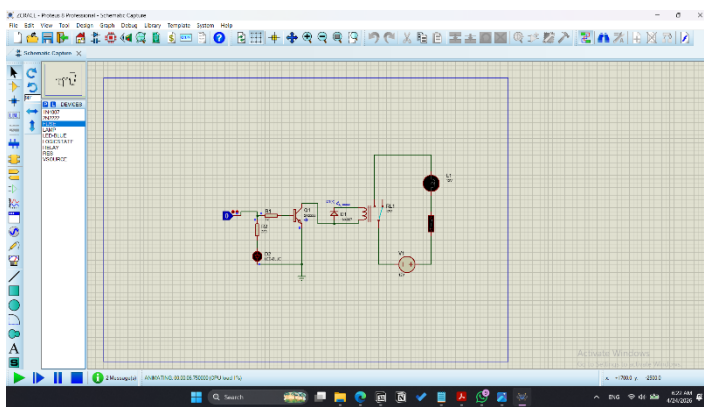
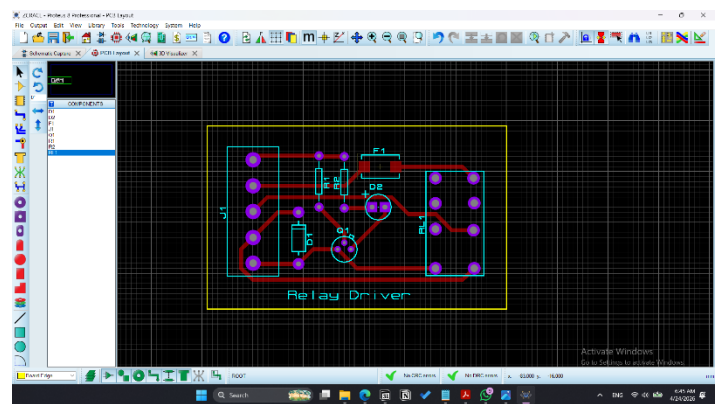
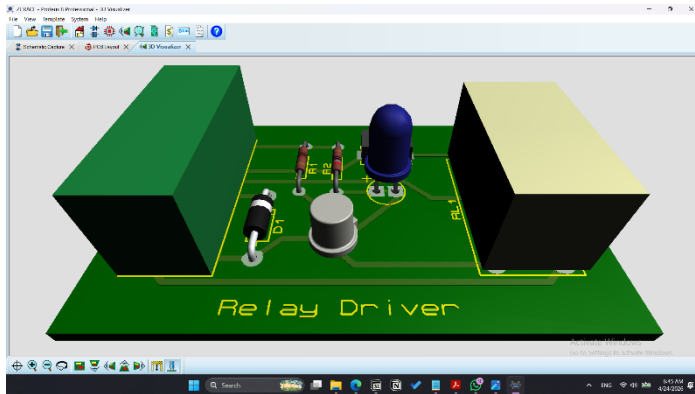
- why a flyback diode is used across the relay coil

To protect the relay when the coil releases its energy the flyback diode works as the path through which this happens which prevents arcs from happening over the coil terminals which can damage the driver circuit.

- what happens if a short circuit occurs in the branch

The first components to react should be the fuse which will sense the high current and open which will protect other parts of the circuit

Images



References

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